Angara Sai Shanmukhi

es24btech11008@iith.ac.in | saishanmukhiangara@gmail.com | +91 9854688999 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.54
TIRUMALA IIT & MEDICAL ACADEMY	Class XII (BIEAP)	2024	98.20%
DR KKRS GOWTHAM EM HIGH SCHOOL	Class X (SSC)	2022	97.33%

PROJECTS

- **GigScore - Alternative Credit Scoring for Gig Workers (JavaScript)**

1. Built a from-scratch, explainable credit-scoring engine using **four custom algorithms -- coefficient of variation, linear regression, weighted moving average, and payment-streak analysis,** to score gig workers' credit worthiness from income and activity patterns instead of formal credit history.
2. Implemented a fraud-detection tool using **Levenshtein edit distance (built via dynamic programming)** to flag typosquatted lender names, combined with a weighted red-flag checklist for common scam signals.
3. Designed a **fully client-side, dependency-free web app** with live score recalculation on editable income data, requiring no backend, database, or build step.

- **Nearest Neighbor Search using Generalized Hyperplane Tree (C++)**

1. Developed a C++ implementation of a **Generalized Hyperplane (GH) Tree** for efficient nearest neighbor search in multidimensional datasets using recursive hyperplane-based partitioning.
2. Implemented **Euclidean distance computation and triangle inequality pruning** to optimize nearest neighbor search, achieving **O(log n) average-case query time**.
3. Improved the application through **modular code design using smart pointers**, input validation, and support for multiple nearest neighbor queries without rebuilding the tree.

- **Minesweeper Game (C++)**

1. Developed a console-based Minesweeper game in C++ with multiple difficulty levels, **randomized mine generation using mt19937**, and an interactive terminal interface using STL.
2. Implemented **safe first-click logic, recursive flood-fill (DFS), and flag/unflag functionality** to accurately replicate core Minesweeper gameplay mechanics.
3. Improved usability through **color-coded console output, comprehensive input validation, and automatic game state management** with win and loss detection.

SKILLS

- **Programming languages :** C, C++,SQL
- **Core Concepts :** Data Structures & Algorithms, Object-Oriented Programming (OOP)
- **Tools :** Git, GitHub, VS Code, Solid Edge

- **Web Development :** HTML, CSS, Java Script

- **Libraries :** STL

RELEVANT COURSES

- **Computer Science Courses :** Introduction to Programming, Data Structures and Applications, Algorithms, Theory of Computation, DBMS-1, Operating Systems-1
- **Mathematics Courses :** Calculus, Elementary Linear Algebra, Differential equations, Numerical Methods, Probability and Random Variables, Discrete Mathematics, Complex Variables

POSITIONS OF RESPONSIBILITY

- **Internship Coordinator, Office of Career Services** *May 2026 - Present*

Assisting with pitching and coordinating with companies for scheduling online assessments and interviews for recruitments.

- **Sponsorship Coordinator, Elan & nVision** *June 2025 - April 2026*

Managed the sponsorship outreach process by researching potential partners, pitching proposals, coordinating branding and marketing deliverables, and ensuring timely communication and follow-ups.

- **Interview and Hosting Head, Media Council** *June 2026 - Present*

Lead event hosting and interview operations by coordinating hosts, conducting interviews, supporting member recruitment, and ensuring the seamless execution of club events and selection processes.

- **Sunshine Mentor, IIT Hyderabad** *June 2025 - April 2026*

Mentored a group of students by providing academic guidance, personal support and organizing sessions to help them adapt smoothly to campus life.

- **Core Member, Shuffle** *June 2025 - April 2026*

Assisted in planning and conducting dance practice sessions, choreographing performances, and coordinating team activities for successful event execution.

EXTRACURRICULAR

- Represented IIT Hyderabad as a performer in the dance team at Inter IIT Cultural 8.0

- Selected for **NSO** (National Sports Organization) **Badminton ,** IIT Hyderabad